MULTI-CANCER EARLY DETECTION (MCED) LIQUID BIOPSY REPORT

Circulating Cell-Free DNA (cfDNA) Methylation & Targeted Somatic Mutation Profiling

PATIENT IDENTIFICATION

Legal Name: Arthur Pendelton Hayes
Preferred Name: Art Hayes
DOB: 1961-04-14 (Age: 65)
Gender: Male
MRN: MRN-7840192

SPECIMEN & ORDER INFORMATION

Accession ID: ACC-2026-MCED-9184
Ordering Physician: Dr. Evelyn Montgomery, MD (NPI: 1487920153)
Ordering Facility: Highland Comprehensive Cancer & Prevention Pavilion
Facility Address: 450 Medical Arts Pavilion, Suite 320, Boston, MA 02115
Collection Date: 2026-08-14 08:30 EDT
Received Date: 2026-08-15 09:15 EDT
Report Date: 2026-08-18 14:00 EDT
Specimen: Whole Blood (Streck cfDNA BCT, 20.0 mL)

OVERALL CANCER SIGNAL STATUS: **SIGNAL DETECTED**

Cancer Signal Origin (CSO 1): Colorectal (84% Probability)
Secondary Origin (CSO 2): Upper GI / Gastric (11% Probability)
Estimated Tumor Fraction: 3.8% ctDNA

SECTION 1: TARGETED SOMATIC GENOMIC ALTERATIONS (NEXT-GENERATION SEQUENCING)

Gene	Genomic Variant / Protein Change	Exon	VAF (%)	Classification / Clinical Tier
KRAS	c.35G>A (p.Gly12Asp / G12D)	Exon 2	2.4%	Tier I - Pathogenic (Colorectal / GI Lineage)
APC	c.4348C>T (p.Arg1450Ter / R1450*)	Exon 16	3.1%	Tier I - Pathogenic (Truncating / Wnt Pathway)
TP53	c.818G>A (p.Arg273His / R273H)	Exon 8	1.9%	Tier I - Pathogenic (DNA Binding Domain)
PIK3CA	Not Detected (Negative for hotspot mutations)	N/A	< 0.1%	Negative
BRAF	Not Detected (Negative for V600E / codon 600)	N/A	< 0.1%	Negative

SECTION 2: EPIGENETIC METHYLATION PATTERNS & CIRCULATING PROTEIN MARKERS

Biomarker Assay	Observed Value	Reference Baseline	Status	Clinical Correlation
SEPT9 Promoter Methylation Index	**0.79**	< 0.15	**ELEVATED**	Strong lower gastrointestinal hypermethylation signal
SDC2 Promoter Methylation Index	**0.68**	< 0.10	**ELEVATED**	Syndecan-2 promoter aberrant epigenetic silencing
NDRG4 Promoter Methylation Index	**0.72**	< 0.12	**ELEVATED**	Colorectal neoplasia associated epigenetic marker
Serum Carcinoembryonic Antigen (CEA)	**12.4 ng/mL**	0.0 - 3.0 ng/mL	**ELEVATED**	Circulating oncofetal glycoprotein antigen elevation
Carbohydrate Antigen 19-9 (CA 19-9)	**48.2 U/mL**	0.0 - 35.0 U/mL	**ELEVATED**	Gastrointestinal / pancreaticobiliary complementary marker
Alpha-Fetoprotein (AFP)	**4.1 ng/mL**	0.0 - 8.0 ng/mL	**NORMAL**	Within reference range; no hepatic oncogenic elevation

SECTION 3: CLINICAL INTERPRETATION & ACTIONABLE RECOMMENDATIONS

CLINICAL SUMMARY: This multi-cancer early detection liquid biopsy assay demonstrates a positive circulating cell-free DNA cancer signal (estimated tumor fraction: 3.8% ctDNA). Epigenetic whole-genome targeted bisulfite methylation mapping reveals pronounced hypermethylation across colorectal-specific locus panels (SEPT9 Index: 0.79, SDC2 Index: 0.68, NDRG4 Index: 0.72), paired with Tier I oncogenic somatic driver mutations (KRAS c.35G>A / p.G12D at 2.4% VAF, APC c.4348C>T / p.R1450* at 3.1% VAF, and TP53 c.818G>A / p.R273H at 1.9% VAF). Together with an elevated serum CEA level of 12.4 ng/mL and elevated CA 19-9 of 48.2 U/mL, the classifier predicts a primary cancer signal origin in the **Colorectal tract** with 84% probability, and a secondary potential signal in the upper GI tract with 11% probability.

RECOMMENDED CLINICAL ACTION: Urgent comprehensive diagnostic evaluation including complete diagnostic colonoscopy with tissue biopsy of any identified mucosal lesions. Consider baseline contrast-enhanced CT of the abdomen and pelvis and multidisciplinary oncology consultation as clinically indicated.

SECTION 4: ASSAY METHODOLOGY & QUALITY CONTROL METRICS

Assay Platform: Targeted bisulfite cfDNA methylation deep sequencing and high-sensitivity somatic mutation panel (Illumina NovaSeq 6000, 150bp paired-end).
Mean Sequencing Depth: 1,250x across targeted loci | **Total cfDNA Input:** 32.5 ng | **Bisulfite Conversion Efficiency:** 99.82% | **Q30 Score:** 94.6%
Limit of Detection (LOD): 0.05% VAF for SNVs/Indels; 0.02% tumor fraction for methylation classifier.
Specimen Quality Control: PASSED (Genomic DNA contamination < 0.01%, hemolysis index negative).

SECTION 5: TEST LIMITATIONS & REGULATORY STATEMENTS

This laboratory-developed test (LDT) was developed and its performance characteristics determined by Vanguard Precision Diagnostics pursuant to CLIA certification regulations (42 CFR 493). This test has not been cleared or approved by the US FDA. The results of this test are intended for use in conjunction with other clinical, radiographic, and histological findings. A 'Signal Detected' result is not an absolute diagnosis of cancer and requires confirmatory diagnostic evaluation by a licensed healthcare provider.

Laboratory Director & Molecular Pathologist:
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Electronic Signature Verification:
Digitally Signed: 2026-08-18 14:12:08 EDT
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